

PSP Oracle Exit - Monolith Database Migration Tiptoe 1 Runbook (Postgres Hash)

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Goal

- Validate Deployment, App and Database Infrastructure built for Postgres Migration.
- Understand and learn the system behaviour functional/non functional in the Postgres database.
- Validate Application Performance and Database Performance/Stability.
- Validate 50% of P0 OLTP workflows and 35% of OLAP workflows.
- Validate the system under less load i.e. Weekend Traffic will have minimal impact.
- Validate Planned Rollback to prepare for any emergency Rollback.
- Validate all the above goals with the minimal customer and partner team impacts.

Tiptoe Summary

Status

Environment	Status
Production	COMPLETED

Tiptoe Progress

JIRA	PSP-28285 - Getting issue details... STATUS
Slack Channel	#psp-monolith-postgres-hash-prod-tiptoe

Duration

Tiptoe Start Time	Tiptoe End Time	Duration
14 Oct 2023 1PM IST	15 Oct 2023 3PM IST	~24 Hours

Tiptoe Highlights

Snoozing A2i

S. No	Application	A2i Maintenance URL
1	DT-Payroll-Svc	https://devportal.intuit.com/app/dp/resource/6847740825712235356/incidentManagement
2	QBDT-Payroll-Service	https://devportal.intuit.com/app/dp/resource/7999536430416583774/incidentManagement
3	CEP	https://devportal.intuit.com/app/dp/resource/6343337667453040796/incidentManagement
4	Case 360	https://devportal.intuit.com/app/dp/resource/1587536460949765378/incidentManagement
5	EWS	https://devportal.intuit.com/app/dp/resource/8649180676662812116/incidentManagement
6	PTC	https://devportal.intuit.com/app/dp/resource/3315792817950450305/incidentManagement
7	QBDT Payroll Entitlement	https://devportal.intuit.com/app/dp/resource/6990730113881900139/incidentManagement (Not onboarded to A2i)
8	Workforce	https://devportal.intuit.com/app/dp/resource/2956630747670426490/incidentManagement
9	BRM	https://devportal.intuit.com/app/dp/resource/3893379470159715175/incidentManagement (Not onboarded to A2i)
10	Desktop Pay API	https://devportal.intuit.com/app/dp/resource/1730525749114265769/incidentManagement
11	Payroll DM	https://devportal.intuit.com/app/dp/resource/8866479358728757267/incidentManagement (Not onboarded to A2i since its not behind gateway)
12	Pubsub	https://devportal.intuit.com/app/dp/resource/651913638447486453/incidentManagement (Not onboarded to A2i since its not behind gateway)
13	EWS Service	https://devportal.intuit.com/app/dp/resource/4901059886671999792/incidentManagement
14	EWS-Rest-IKS Service	https://devportal.intuit.com/app/dp/resource/6270154173504901366/incidentManagement

Downtime Details

App	Tiptoe (PST)			Rollback (PST)		
	Downtime Start	Downtime End	Actual	Downtime Start	Downtime End	Actual
PSP	10/14/2023 00:43 AM	10/14/2023 01:50 AM	1 hr 7 mins	10/14/2023 10:39PM	10/14/2023 11:22PM	43min
PAPI	10/14/2023 00:36 AM	10/14/2023 01:51 AM	1 hr 15 mins	10/14/2023 10:37PM	10/14/2023 11:34PM	57min
Pubsub	10/14/2023 00:36 AM	10/14/2023 01:51 AM	1 hr 15 mins	10/14/2023 10:37PM	10/14/2023 11:30PM	53min
RJF	10/14/2023 00:36 AM	10/14/2023 01:51 AM	1 hr 15 mins	10/14/2023 10:37PM	10/14/2023 11:33PM	56min
PayrollDM	10/14/2023 00:36 AM	10/14/2023 01:51 AM	1 hr 15 mins	10/14/2023 10:37PM	10/14/2023 11:32PM	55min

Downtime Summary

Downtime	Estimated	Actual	Tiptoe/Rollback Window		Comments
			Start Time (PST)	End Time (PST)	
Tiptoe	105 Mins	1 hr 15 mins	10/14/2023 00:36 AM	10/14/2023 01:51 AM	Time Split Expected 1 hour 15 mins of Deploying all apps at full scale. 30 Mins for database side steps - Database Replication Lag is 0. - Stopping A B and creating B A replication tasks or vice versa in case of rollback.
Planned Rollback	120 Mins				Time Split Expected - Database Replication Lag is 0. 15 Mins for Sanity.
Emergency Rollback (In case of issues during the Tiptoe window)	170 Mins				Time Split Expected 120 Mins for Planned Rollback 30 Mins Debugging and Replication Lag to settle down. 15 Mins of additional time for Sanity.

Learnings/Enhancements

S No	Learning	Action Item	Owner	Priority	Status	Comments
1						
2						
3						
4						

Important Resources

Dev Portal

Application	Dev Portal Link
PSP	PSP Dev Portal
PAPI	PAPI Dev Portal
PubSub	Pubsub Dev Portal
RJF	RJF Dev Portal
PayrollDM	PayrollDM Dev Portal

JSS

Application	JSS Link
PSP Batch Jobs	PSP JSS View
RJF Reports	RJF JSS View

Splunk

Type	Link
Application	Splunk Application Production

Gateway	Splunk Gateway Production
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ArgoCd

Environment	Application	Dev Portal Link
Production	PSP	Production PSP ArgoCd
	PAPI	Production PAPI ArgoCd
	Pubsub	Production Pubsub ArgoCd
	RJF	Production RJF ArgoCd
	PayrollDM	Production PayrollDM ArgoCd

Notification Email Addresses

i SBSEG-PD QBDT PSP AWS <SBSEG-PDQBDTPSPAWS@intuit.com>; SBG PSP Services Decomp Scrum <SBGPSPServicesDecompScrum@intuit.com>; SBSEG SBG Payroll PSP Informed <SBGPayrollPSPInformed@intuit.com>; SBSEG-Iqbal-Extended <SBSEG-PDIqbalExtended@intuit.com>; SBSEG-PD DTPayrollExtended <SBSEG-PDDTPayrollExtended@intuit.com>; SBSEG-PD DTPayroll <SBSEG-PDDTPayroll@intuit.com> Kaura, Varun <Varun_Kaura@intuit.com>; Narlagalla, Praveen Goud <PraveenGoud_Narlagalla@intuit.com>; Swain, Amrit <Amrit_Swain@intuit.com>; Johnson, Anu <Anu_Johnson@intuit.com>, anuradha_kumar i@intuit.com, rohan_rasam@intuit.com, narendraprasad_nanna@intuit.com, subulakshmi_nagarajan@intuit.com, rajeevkumar_singh@intuit.com, ketan_popat@intuit.com, Pahuja, Hitesh <Hitesh_Pahuja@intuit.com>, Huang, Alex <Alex_Huang@intuit.com>

Change Requests

Application	Tiptoe			Rollback		
	Change Request	Deployment Image	Status	Change Request	Deployment Image	Status
PSP	CHG2564403	master-eee3694	APPROVED	CHG2564502	master-eee3694	APPROVED
PAPI	CHG2564465	master-2df7a15	APPROVED	CHG2564515	master-2df7a15	APPROVED
Pubsub	CHG2564491	master-4fccdba	APPROVED	CHG2564518	master-4fccdba	APPROVED
RJF	CHG2564494	master-cac9e26	APPROVED	CHG2564520	master-cac9e26 / master-693e09d	APPROVED
PayrollDM	CHG2564498	master-ed59a24	APPROVED	CHG2564531	master-ed59a24 / master-6fda8f0	APPROVED

Config Map Pull requests

TipToe Pull Requests

Application	Pull Requests
PSP	
PAPI	
Pubsub	
RJF	

Rollback Pull Requests

Application	Pull Requests
PSP	
PAPI	

Pubsub	
RJF	

Runbook

Pre Tiptoe Steps

S No	Task	Steps	Due Date	Status	Owner								
1	Partner Sign-off.	Refer NotificationEmailAddresses Section. Partner Contacts : Partner Team	12 Oct 2023	DONE	Nagarajan, Subulakshmi Nanna, Narendra Prasad								
2	Inform partner teams slack channel.	Ask the stake holders to be prepared for the downtime. Inform following Slack Channels. #payroll-pd-releases, #ioc, #psp-rtb-priority, #sbg-payroll-prod, #mse-commerce-support, #cx3-production	12 Oct 2023	DONE	Bhat, Chethan								
3	Send Release Freeze Email to Dev Team.	No Releases in PSP / PAPI / Pubsub / RJF allowed in this interval.	12 Oct 2023	DONE	N, Ramasubramani								
4	Care Sign-off	SAP can be validated in Staging Environment.	12 Oct 2023	NA	Sur, Koustav								
5	Inform Leadership Team on the Risk & Downtime	Inform Leadership Team on the Risk & Downtime.	12 Oct 2023	DONE	Nasim, Syed Iqbal								
6	Prepare QBDT Production Setup.	Prepare QBDT Production Setup.	12 Oct 2023	DONE	Swain, Amrit Hoolimath, Praveenkumar								
7	Schema object count check	Count of objects are matching except known ones for partitioning.	12 Oct 2023	DONE	Choubey, Mayank								
8	Time for Validating Latest data during the Tiptoe window	Unknown User (rrasam) / Rajoria, Himanshu to update expected numbers and veridata url and job names. <table><tr><th>Activity</th><th>Job Name</th><th>Expected</th><th>Actual</th></tr><tr><td>Tiptoe</td><td></td><td>35 mins</td><td>35 mins</td></tr></table>	Activity	Job Name	Expected	Actual	Tiptoe		35 mins	35 mins	12 Oct 2023	DONE	Rajoria, Himanshu / Unknown User (rrasam)
Activity	Job Name	Expected	Actual										
Tiptoe		35 mins	35 mins										
9	Decide strategy for Monthly & Quarterly Jobs.	Decide strategy for Monthly & Quarterly Jobs.	12 Oct 2023	NA	Swain, Amrit								
10	Make sure there is no IKS Cluster Upgrade during the Tiptoe window.	- Talk to MSaaS team and confirm. - Get the contacts of MSaaS team in case if we need assistance during tiptoe. - MSaaS Signoff - msaas-signoff	12 Oct 2023	DONE	Bansal, Jatin								
11	Make sure there is no Splunk Maintenance during the Tiptoe window.	Slack Channel : #pi-tools-splunk-spt Splunk Signoff : splunk-signoff	12 Oct 2023	DONE	Bansal, Jatin								
12	We use PRs for the release. Make sure there is no Git maintenance during the Tiptoe window.	- Talk to Git team and confirm. - Git Signoff - git-signoff	12 Oct 2023	DONE	Bansal, Jatin								
13	Capture 07 Oct 2023 - 08 Oct 2023 traffic for each app.	- Update AppsHealth section with the last week traffic. - Post the traffic and health numbers in tiptoe channel.	12 Oct 2023	DONE	Singh, Sanket Unknown User (akumari3) Goswami, Harshita Singhal, Vikas								

14	Make sure the last Master branch deployed did not have any deployment errors.	Look for errors in this dashboard	12 Oct 2023	DONE	Singhal, Vikas																		
		<table><tr><th>App</th><th>Dashboard</th><th>Status</th></tr><tr><td>PSP</td><td>PSP Unknown Errors</td><td> DONE </td></tr><tr><td>PAPI</td><td>PAPI Unknown Errors</td><td> DONE </td></tr><tr><td>Pubsub</td><td>PubSub Unknown Errors</td><td> DONE </td></tr><tr><td>RJF</td><td>RJF Unknown Errors</td><td> DONE </td></tr><tr><td>Payroll DM</td><td>PayrollDM Unknown Errors</td><td> DONE </td></tr></table>	App	Dashboard	Status	PSP	PSP Unknown Errors	DONE	PAPI	PAPI Unknown Errors	DONE	Pubsub	PubSub Unknown Errors	DONE	RJF	RJF Unknown Errors	DONE	Payroll DM	PayrollDM Unknown Errors	DONE			
App	Dashboard	Status																					
PSP	PSP Unknown Errors	DONE																					
PAPI	PAPI Unknown Errors	DONE																					
Pubsub	PubSub Unknown Errors	DONE																					
RJF	RJF Unknown Errors	DONE																					
Payroll DM	PayrollDM Unknown Errors	DONE																					
15	Meet to decide GO/No GO		12 Oct 2023	DONE	N, Ramasubramani Muthurangam, Karthikeyan Nasim, Syed Iqbal Choubey, Mayank Bhat, Chethan																		
16	Send Downtime Update to stake holders via Email.	Refer NotificationEmailAddresses Section. Ask the stake holders to be prepared for the downtime.	12 Oct 2023	DONE	Bhat, Chethan																		
17	Reassign PagerDuty.	1. Update PagerDuty Schedule for the Tiptoe window. 2. Update PagerDuty Schedule for the next week after the Tiptoe. Pager Duty Schedules QBPY - DPS : EMS OPS RNO Primary SBG - QBPY - DPS : DevOps PSP (IDC)	12 Oct 2023	DONE	Bhat, Chethan																		
18	Note down all the PSP JSS Jobs in the same window.	Link : PSP JSS View Update Failed/MissedBatchJobs Section.	12 Oct 2023	DONE	Karra, Vineeth Hoolimath, Praveenkumar																		
19	Note down all the RJF JSS Jobs in the same window.	Link : RJF JSS View Update Failed/MissedBatchJobs Section.	12 Oct 2023	DONE	Hoolimath, Praveenkumar Karra, Vineeth																		
20	IXP flag updates	Update ENABLE_RUNTIME_HINT and RUNTIME_HINT_VALUES IXP flag	14 Oct 2023	DONE	Goswami, Harshita																		
21	Update IDPS Postgres Hash Credentials.	Steps: Pre-requisite: install StashCopy .pem file to /tmp folder. eiamcli login - Add Secrets from Command Line. <code>sh Oracle-Hash-A.sh -a</code> <code>sh Postgres-Hash-B.sh -a</code> <code>sh Oracle-Hash-C.sh -a</code> - Validate Secrets from Command Line. <code>sh Oracle-Hash-A.sh -v</code> <code>sh Postgres-Hash-B.sh -v</code> <code>sh Oracle-Hash-C.sh -v</code> - Validate from Developer Portal	12 Oct 2023	DONE	Bhagat, Suyash																		
22	Update CR and Image Details	Create Change Requests & decide images for Tiptoe and Rollback. Refer ChangeRequests Section	12 Oct 2023	DONE	Unknown User (mvenkatesh)																		
23	Bring down staging and middleware apps.	- Bring down staging and middleware apps. - Ensure no database connections from Middleware and Staging. - Bringdown the databases.	12 Oct 2023	DONE	Bahl, Pulkit Choubey, Mayank																		
24	Ensure there is no OOS or any data issue between Oracle Hash and Postgres Hash	- Veridata reports should not have any OOS. - Ensure latest/last 9 days and 1 day compare is green. Veridata Console - Job Name:	12 Oct 2023	DONE	Unknown User (rrasam) / Rajoria, Himanshu																		

25	Run Sanity for PSP/PAPI/Pubsub/RJF. Execute the sanity at 9AM on the Tiptoe day.	Execute below Sanity Jobs to ensure jobs working fine as expected. <table><tr><th>App</th><th>Sanity Link</th><th>Status</th></tr><tr><td>PSP</td><td>PSP Sanity</td><td>DONE</td></tr><tr><td>PAPI</td><td>PAPI Sanity</td><td>DONE</td></tr><tr><td>Pubsub</td><td>Pubsub Sanity</td><td>DONE</td></tr><tr><td>RJF</td><td>RJF Sanity</td><td>DONE</td></tr><tr><td>PayrollDM</td><td>PayrollDM Sanity</td><td>DONE</td></tr></table>	App	Sanity Link	Status	PSP	PSP Sanity	DONE	PAPI	PAPI Sanity	DONE	Pubsub	Pubsub Sanity	DONE	RJF	RJF Sanity	DONE	PayrollDM	PayrollDM Sanity	DONE	12 Oct 2023	DONE	Goswami, Harshita
App	Sanity Link	Status																					
PSP	PSP Sanity	DONE																					
PAPI	PAPI Sanity	DONE																					
Pubsub	Pubsub Sanity	DONE																					
RJF	RJF Sanity	DONE																					
PayrollDM	PayrollDM Sanity	DONE																					
26	Validate PSP Node Level Sanity.	Validate Node level Sanity and Dashboard Node Level Sanity Dashboard .	12 Oct 2023	DONE	Goswami, Harshita																		
27	Update Security Group to add Production Connection	Update Security Group to add Production Connection	12 Oct 2023	DONE	Choubey, Mayank																		
28	Integration Test Suite	https://build.intuit.com/intuit-desktop-payroll/job/iks-intg/4798/ Integration-report(no 4798).pdf	13 Oct 2023	DONE	Unknown User (mvenkatesh)																		
29	Automation Test Suite	PSP - https://build.intuit.com/intuit-desktop-payroll/job/payroll-automation-suite-east-e2e/261/cucumber-html-reports/overview-features.html - PSP Automation Reports (no 261).pdf https://build.intuit.com/intuit-desktop-payroll/job/payroll-automation-suite-east-e2e/263/cucumber-html-reports/overview-features.html - PSP Automation Reports (no 263).pdf PAPI https://build.intuit.com/intuit-desktop-payroll/job/payroll-automation-suite-east-e2e/262/cucumber-html-reports/overview-features.html - PAPI Automation Repport (no 262).pdf	13 Oct 2023	DONE	Unknown User (mvenkatesh)																		
30	Team to validate Database connectivity for A/B/C Prod instances.	Choubey, Mayank to create a read-only user on A/B /C instances. - Team to validate Database connectivity for A/B/C Prod instances.	12 Oct 2023	DONE	Choubey, Mayank Bhagat, Suyash Goswami, Harshita																		
31	Update Maintenance Window	Update Maintenance Window. Refer Maintenance Windows section. - Sunday (10AM - 1PM) IST - Monday (10AM - 1PM) IST Refer below threads and confirm they snoozed their asset as well. MSE Team CX3 Team	12 Oct 2023	DONE	Bhagat, Suyash Singh, Sanket																		
32	Create PR to update Config Map in master deployment branch and update the pull request section		12 Oct 2023	NA	Singh, Sanket																		
33	Run/Pause the Jobs	- Run the LedgerBalance Job after 12:30PM IST - Run the State report processor after 12:30PM IST - Pause MtlTransactionReportEnrichProcessor - Pause SoxReport - Pause SoxDBReport - Pause RTBAutomationJob	14 Oct 2023	DONE	Hoolimath, Praveenkumar																		
34	Validate Pod Postgres connectivity	Create a dummy pod and validate the connectivity to Postgres reader instances	13 Oct 2023	DONE	Joshi, Vishwanath																		

TipToe Steps

S No	Task	Steps	Status	Owner	Remarks														
1	Update Maintenance Window	Update Maintenance Window. Refer Maintenance Windows section. - Sunday (10AM - 1PM) IST - Monday (10AM - 1PM) IST Refer below threads and confirm they snoozed their asset as well. MSE Team CX3 Team	DONE	Bhagat, Suyash Singh, Sanket															
2	Data Parity	Ensure latest 1 day compare and sanity check output is green.	OPEN	Unknown User (rrasam) Rajoria, Himanshu															
3	Send Downtime Update to stake holders via Email.	CR, Apps & Downtime details to be shared. Refer ChangeRequests Section. Refer NotificationEmailAddresses Section.	OPEN	Bhat, Chethan															
4	Specific communication to #ioc Nasim, Syed Iqbal	Inform #ioc team on ICP services not being snoozed and leading and it leading to incident. This is required because of Tax HCAP and ICP team dont want to miss the alerts.	OPEN	Nasim, Syed Iqbal															
5	Inform partner teams slack channel.	Ask the stake holders to be prepared for the downtime. CR, Apps & Downtime details to be shared. Refer ChangeRequests Section. Inform following Slack Channels. #payroll-pd-releases, #ioc, #psp-rtb-priority, #sbg-payroll-prod, #mse-commerce-support, #cx3-production payroll-pd-releases ioc psp-rtb-priority sbg-payroll-prod mse-commerce-support cx3-production	OPEN	Bhat, Chethan															
6	Ensure there are no critical jobs running.	- Navigate to JSS View Link. - Wait for critical jobs to complete.	OPEN	Hoolimath, Praveenkumar Karra, Vineeth															
7	Bring down below apps. 1. Pubsub 2. RJF 3. PAPI 4. PayrollDM	Bring down Pubsub, PAPI, RJF. Zero Pods With Maintenance Page	OPEN	Goswami, Harshita															
8	Bring down PSP along with Maintenance Page	Bring down PSP with Maintenance Page. Zero Pods PSP With Maintenance Page	OPEN	Goswami, Harshita															
9	Ensure all apps are down.	Hit following URLs for each application. <table><tr><th>App</th><th>Health Check URL</th></tr><tr><td>PSP</td><td>sh PSPHealthCheck.sh prd</td></tr><tr><td>SAP</td><td>SAP</td></tr><tr><td>PAPI</td><td>PAPI Health Check</td></tr><tr><td>Pubsub</td><td>Pubsub Sanity</td></tr><tr><td>RJF</td><td>RJF Health Check</td></tr><tr><td>PayrollDM</td><td>PayrollDM Health Check</td></tr></table>	App	Health Check URL	PSP	sh PSPHealthCheck.sh prd	SAP	SAP	PAPI	PAPI Health Check	Pubsub	Pubsub Sanity	RJF	RJF Health Check	PayrollDM	PayrollDM Health Check	OPEN	Rajvanshi, Venkat Singh, Sanket	
App	Health Check URL																		
PSP	sh PSPHealthCheck.sh prd																		
SAP	SAP																		
PAPI	PAPI Health Check																		
Pubsub	Pubsub Sanity																		
RJF	RJF Health Check																		
PayrollDM	PayrollDM Health Check																		

10	Update the downtime details section for all the apps.	Update DowntimeDetails Section.	OPEN	Bhat, Chethan	
11	Note down All the failed RJF /PSP BatchJobs.	Update Failed/MissedBatchJobs Section.	OPEN	Hoolimat h, Praveenk umar Karra, Vineeth	
12	Lock application users on Oracle Hash db.	Choubey, Mayank to update all database steps • Wait for Inflight requests/Running Batch Jobs to complete. • There should not be any connections to Oracle Hash Database. • Lock Oracle Monolith Users : PSPADM and PSPAPP <pre> alter user PSPADM account lock; alter user PSPAPP account lock; alter user PSPREAD account lock; alter user PSPREAD_AWS account lock; select username,account_status from dba_users where username in ('PSPAPP', 'PSPADM', 'PSPREAD', 'PSPREAD_AWS'); </pre>	OPEN	Choubey, Mayank	
13	Ensure there are no active connections to the pre-production Oracle Hash database.	No active connections from PSP / PAPI / Pubsub / RJF. Monolith and Monolith ReadReplica : <pre> set lines 300 col username for a30 col machine form a70; select username,machine,count(*) from gv\$session where username is not NULL and username not in ('RDSADMIN','INTUADMIN','SYS','SYSTEM') and type = 'USER' group by service_name, username,machine order by username,machine; </pre> (Kill Connections if needed) Monolith <pre> select 'exec rdsadmin.rdsadmin_util.kill ' s.sid ',' s. serial# ',' ''''PROCESS'''' ';' from gv\$session s where USERNAME in ('PSPAPP','PSPADM'); </pre>	OPEN	Choubey, Mayank	
14	Check replication lag	check replication lag for C2->D2, C2D2'	OPEN	Choubey, Mayank	
15	Ensure replication lag is zero	Wait till Oracle Hash (B) Postgres Hash(C2) Replication lag to zero.	OPEN	Choubey, Mayank	

16	Create and alter sequences	Create Sequences on C2 and alter sequences to start with last_number from oracle +1 <pre>SELECT 'ALTER SEQUENCE ' sequence_owner '. ' sequence_name ' RESTART WITH ' (last_number + 1) ';' FROM all_sequences WHERE sequence_owner = 'PSPADM';</pre> <pre>\i CreateSequence.sql \i seq_alter_seq_onC2.sql</pre>	OPEN	Choubey, Mayank	
17	Enable Triggers	Enable Triggers on C2. login to C2 host : ggpspadm01.sbg-ppsp-prod.a.intuit.com <pre>\i CreateTrigger.sql</pre>	OPEN	Choubey, Mayank	
18	Disable FK Constraints & triggers on Oracle Hash	Disable all FKs & triggers on B after setting settag=2 host : ggpspadm01.sbg-ppsp-prod.a.intuit.com <pre>@disable_triggers_on_B.sql @disable_constraints_on_B.sql</pre> Validation count is 0 for both of below <pre>select constraint_name, table_name,status from dba_constraints where owner='PSPADM' and constraint_type='R' and status='ENABLED';</pre> <pre>select trigger_name,table_name,status from dba_triggers where owner='PSPADM' and status='ENABLED';</pre>	OPEN	Choubey, Mayank	
19	Create & Start Extract on C2	Hostname - ggpspag01.sbg-ppsp-prod.a.intuit.com Register Extract <pre>obey ./diroby/dblogin_c2.oby register extract EPP1PG02 with database pspag02</pre> Add Extract <pre>obey ./diroby/create_epp1pg02.oby</pre> Test with insert on heartbeat table to ensure its extracting data from C2 <pre>insert into pspadm.gg_heartbeat ('C2_to_B', current_timestamp);</pre> <pre>stats EPP1PG02</pre>	OPEN	Choubey, Mayank	

20	Create pump for C2 to B	Hostname - ggpsppag01.sbg-ppsp-prod.a.intuit.com <pre>obey ./diroby/create_ppsphp02.oby info ppsphp02 start ppsphp02 stats ppsphp02</pre>	OPEN	Choubey, Mayank	
21	Create & Start Replicat for C2->B.	Create & Start Replicat for C2->B. (settag=2, and during downtime) Hostname - ggpsphpp05.sbg-ppsp-prod.a.intuit.com Add the Relicats <pre>obey ./diroby/create_replicats.oby obey ./diroby/create_coordinated_replicats.oby</pre> Register the Replicats <pre>dblogin useridalias ggtarget register replicat rpsppg51 database register replicat rpsppg52 database register replicat rpsppg53 database register replicat rpsppg54 database register replicat rpsppg55 database register replicat rpsppg56 database register replicat rpsppg57 database register replicat rpsppg58 database</pre>	OPEN	Choubey, Mayank	
22	Validate replication C2->B using heartbeat table	Validate replication C2->B using heartbeat table Validate replication C2->D2 using heartbeat table <pre>stop RPPG3 start RPPG3</pre>	OPEN	Choubey, Mayank	
23	Unlock users in Postgres Hash (C2)	<pre>alter user pspapp with login; alter user pspadm_owner with login; alter user pspbatch_ro_user with login; alter user pspbatch_rw_user with login;</pre>	OPEN	Choubey, Mayank	
24	Restart extract on B (for B->C2)	Hostname - ggpsphpp05.sbg-ppsp-prod.a.intuit.com Edit extract parameter file to include TRANLOGOPTIONS FETCHPARTIALLOB <pre>stop EPSPHNP2 edit parameter file epsphnp2.prm start EPSPHNP2</pre>	OPEN	Choubey, Mayank	
25	Validate Production to point to Postgres Hash(C2) Database.	Validate Read Write / VMP / RJF Read Instances (3 instances in total). • Pre-requisite: install Stash • Copy .pem file to /tmp folder. • eiamcli login • Validate IDPS Credentials (Database URL/Username/Password) <i>sh Postgres-Hash-B.sh -v</i>	OPEN	Bhagat, Suyash	

26	Bring up PSP Pods with Maintenance.	Bring up PSP with Maintenance page. Scale up PSP Pods With Maintenance Page. Ensure correct image is deployed. • Dashboard : Validate from DT Payroll Pod Info • Script : imageDetails.sh prd • Validate database in config map	OPEN	Goswami , Harshita											
27	Monitor if PSP connecting to Postgres Hash Database	Monitor Database Connection Dashboard and validate.	OPEN	Bahl, Pulkit											
28	Ensure there are no errors in the startup logs.	Look for errors in this dashboard Unknown Errors	OPEN	Bhagat, Suyash											
29	Check the Node level Sanity Logs.	Node Level Sanity Dashboard - Check if all the pods passed sanity tests.	OPEN	Bahl, Pulkit											
30	Monitor DB Connections.	Monitor db connetions	OPEN	Choubey , Mayank											
31	Monitor Overall database /replication health vi CW dashboard	Postgres Hash Database Performance Monitor GoldenGate Replication Dashboard	OPEN	Choubey , Mayank											
32	Bring up PSP pods without Maintenance.	Bring up PSP without Maintenance page. Scale up PSP Pods Without Maintenance Page. Ensure correct image is deployed. • Dashboard : Validate from DT Payroll Pod Info • Script : imageDetails.sh prd • Validate database in config map	OPEN	Goswami , Harshita											
33	Run the PSP Sanity.	Execute below Sanity Jobs to ensure jobs working fine as expected. <table><tr><th>Application</th><th>Sanity Link</th><th>Status</th></tr><tr><td>PSP</td><td>PSP Sanity</td><td> OPEN </td></tr><tr><td>SAP</td><td>SAP Sanity</td><td> OPEN </td></tr></table>	Application	Sanity Link	Status	PSP	PSP Sanity	OPEN	SAP	SAP Sanity	OPEN	OPEN	Rajvanshi , Venkat Singh, Sanket		
Application	Sanity Link	Status													
PSP	PSP Sanity	OPEN													
SAP	SAP Sanity	OPEN													
34	Run replication sanity (count based table records validation)	sanity-table-rowcount-check.sql	OPEN	Unknown User (rrasam)											
35	Update the downtime details section for PSP.	Update DowntimeDetails Section.	OPEN	Bhat, Chethan											
36	Monitor DB Performance	We will be using combination of Performance Insights, monitoring dashboard and adhoc scripts / slow query log based on need	OPEN	Choubey, Mayank Popat, Ketan Gunvantrai											
37	Bring up all pods for below services 1. PAPI 2. RJF 3. Pubsub 4. PayrollDM	Bring up Pubsub, PAPI and RJF apps without maintenance page. Bring up Apps Without Maintenance Page Ensure correct image is deployed. • Dashboard : Validate from DT Payroll Pod Info • Script : imageDetails.sh prd • Validate database in config map	OPEN	Goswami , Harshita											
38	Monitor if all apps connecting to Postgres Hash Database	Monitor Database Connection Dashboard and validate.	OPEN	Bahl, Pulkit											
39	Ensure there are no errors in the startup logs.	Monitor and Report unknown errors. <table><tr><th>Application</th><th>Dashboard</th></tr><tr><td>PAPI</td><td>PAPI Unknown Errors</td></tr><tr><td>Pubsub</td><td>PubSub Unknown Errors</td></tr><tr><td>RJF</td><td>RJF Unknown Errors</td></tr><tr><td>PayrollDM</td><td>PayrollDM Unknown Errors</td></tr></table>	Application	Dashboard	PAPI	PAPI Unknown Errors	Pubsub	PubSub Unknown Errors	RJF	RJF Unknown Errors	PayrollDM	PayrollDM Unknown Errors	OPEN	Singhal, Vikas	
Application	Dashboard														
PAPI	PAPI Unknown Errors														
Pubsub	PubSub Unknown Errors														
RJF	RJF Unknown Errors														
PayrollDM	PayrollDM Unknown Errors														

40	Run PAPI/Pubsub/RJF /PayrollDM Sanity.	Execute below Sanity Suites. <table><thead><tr><th>Application</th><th>Sanity Link</th><th>Status</th></tr></thead><tbody><tr><td>PAPI</td><td>PAPI Sanity</td><td> OPEN </td></tr><tr><td>Pubsub</td><td>Pubsub Sanity</td><td> OPEN </td></tr><tr><td>RJF</td><td>RJF Sanity</td><td> OPEN </td></tr><tr><td>PayrollDM</td><td>PayrollDM Sanity</td><td> OPEN </td></tr></tbody></table>	Application	Sanity Link	Status	PAPI	PAPI Sanity	OPEN	Pubsub	Pubsub Sanity	OPEN	RJF	RJF Sanity	OPEN	PayrollDM	PayrollDM Sanity	OPEN	OPEN	Rajvanshi , Venkat Singh, Sanket	
Application	Sanity Link	Status																		
PAPI	PAPI Sanity	OPEN																		
Pubsub	Pubsub Sanity	OPEN																		
RJF	RJF Sanity	OPEN																		
PayrollDM	PayrollDM Sanity	OPEN																		
41	Ensure we see DB Connections from Pubsub / PAPI / RJF / PayrollDM.	Monitor db connetions	OPEN	Choubey, Mayank																
42	Update the downtime details section for all the apps.	Update DowntimeDetails Section.	OPEN	Muthuran gam, Karthikey an																
43	Run PSP & RJF failed jobs and missed Jobs.	Refer Failed/MissedBatchJobs Section. Run LedgerBalance Job if Missed.	OPEN	Hoolimat h, Praveenk umar Karra, Vineeth																
44	Monitor all the apps and databases.	Refer Monitoring Section.	OPEN	Everyone																
45	Ensure replication is running without any issues & monitor the lag	GoldenGate Replication Dashboard	OPEN	Choubey, Mayank																
46	Review & Take action on all the pending Alerts.	Pending Alerts - Critical Alerts PD Pending Alerts - RNO PD	OPEN	Bhat, Chethan N, Ramasub ramani																
47	Run Payroll From Production Test company	IKS QBDT PSP Adapters	OPEN	Swain, Amrit																
48	Monitor RJF & PSP Batch Jobs.	- Update JSS Jobs Schedule completed jobs status every 2hours. (validate job using sql queries) - Update RJF Jobs Schedule completed jobs status every 2hours. (validate job with splunk)	OPEN	Hoolimat h, Praveenk umar Karra, Vineeth																
49	Merge generated PRs to environment/prd-usw2-iks to update config map in manifest to point to oracle for PSP Papi Pubsub RJF		OPEN	Bhagat, Suyash																
50	Merge PRs to the master branch to update config map to oracle for PSP Papi Pubsub RJF refer PR section		NA	., Prince																

Post TipToe Steps

S No	Task	Steps	Status	Owner	Remarks
1	Send Downtime Update to stake holders via Email.	Refer NotificationEmailAddresses Section.	OPEN	Bhat, Chethan	

2	Offline Monitoring for all the Apps.	Refer Monitoring Section. Update once in every 2 hours in tiptoe slack channel.	OPEN	All	
3	Review & Take action on all the pending Alerts.	Pending Alerts - Critical Alerts PD Pending Alerts - RNO PD	OPEN	Hoolimath, Praveenkumar Karra, Vineeth	
4	Review & Close all pending CRs.	Refer ChangeRequests Section.	OPEN	Sur, Koustav	
5	Inform partner teams slack channel.	Inform following Slack Channels. #payroll-pd-releases, #ioc, #psp-rtb-priority, #sbg-payroll-prod, #mse-commerce-support, #cx3-production • payroll-pd-releases • ioc • psp-rtb-priority • sbg-payroll-prod • mse-commerce-support • cx3-production	OPEN	Bhat, Chethan	
6	Monitor Critical Batch Jobs.	Monitor Critical Batch Jobs (Required Only for the Cutover) Refer CriticalBatchJobsMonitoring Section.	OPEN	Hoolimath, Praveenkumar	
7	Monitor database Stability, Availability & Performance	We will be using combination of Performance Insights, monitoring dashboard and adhoc scripts / slow query log based on need	OPEN	Choubey, Mayank (Primary) Popat, Ketan Gunvantrai (Escalation)	
8	Monitor replication from Postgres Hash to Oracle Hash	• Monitor DMS replication lag from Postgres Hash to Oracle Hash • Keep running sanity counts. • Ensure Veridata validation is green from Postgres Hash to Oracle Hash	OPEN	Narlagalla, Praveen Goud	
9	Schedule Veridata to validate Delta Data.	Schedule Veridata to validate Delta Data. Veridata Console Link Veridata Job Name(s): PSP_P2O_last_2_days_C2_B PSP_P2O_last_2_days_C2_D2 PSP_P2O_last_2_days_C2_D2Prime	OPEN	Unknown User (rrasam) Rajoria, Himanshu	
10	Apply Repair Scripts if applicable.	Apply Repair Scripts if applicable.	OPEN	Choubey, Mayank	

Rollback Decision and Action Plan

Issue Category	Scenario	Action Item
Application Functionality	Any of the following issues. • P0/P1 customers workflow impacted. • P0/P1 agents workflow impacted. • P0/P1 Batchjobs impacted.	Rollback and Restore application/service.
	P2/P3 customers/agents/BatchJobs issue.	No Rollback. Fix and Move Forward.

Application Performance	Any of the following issues. - P0/P1 customers workflow impacted. - P0/P1 agents workflow impacted. - P0/P1 Batchjobs impacted.	1. Try below steps if this works. - Flush the current plan. - Baseline the plan. 2. If nothing works, if performance degrade is in acceptable range, move forward. 3. If nothing works, if performance degrade is not in acceptable range, Rollback and Restore.
	P2/P3 customers/agents/BatchJobs issue.	Fix and move forward. Try below steps. - Flush the current plan. - Baseline the plan. Stay even if performance is degraded. Pause the jobs if required.
VMP/Equifax	VMP and Equifax performance degraded.	Switch VMP and Equifax traffic from Reader to Read/Write instance.
Replication	Replication task is running into error and replication is broken due to following scenarios.	Replication broken in A and C. Fix and move forward. No rollback is possible here.
Data parity	Data Parity fails	Fix and Forward. Massive Out of Sync in 1 instance. Generate OOS fix script and fix OOS.

Rollback Steps

	Task	Steps	Status	Owner	Remarks
1	Update Maintenance Window	Update Maintenance Window. Refer Maintenance Windows section. - Sunday (10AM - 1PM) IST - Monday (10AM - 1PM) IST Refer below threads and confirm they snoozed their asset as well. MSE Team CX3 Team	DONE	Singh, Sanket	
2	Monitor Postgres Hash --> Oracle Hash Replication lag	Monitor C2B, C2D2, C2->D2 Replication lag and ensure on lag.	DONE	Unknown User (rrasam) R ajoria, Himanshu Choubey, Mayank	
3	Data Parity - Ensure Veridata Validation on Latest Data is complete.	Trigger validation for the Tip-toe period for C2B, C2D2, C2->D2			
4	Note down all the PSP JSS Jobs in the same window.	Link : PSP JSS View Update Failed/MissedBatchJobs Section.	DONE	Karra, Vineeth	
5	Note down all the RJF JSS Jobs in the same window.	Link : RJF JSS View Update Failed/MissedBatchJobs Section.	DONE	Karra, Vineeth	
6	Inform slack channel about the downtime.	Ask the stake holders to be prepared for the downtime. CR, Apps & Downtime details to be shared. Refer ChangeRequests Section. Inform following Slack Channels. #payroll-pd-releases, #ioc, #psp-rtb-priority, #sbg-payroll-prod, #mse-commerce-support, #cx3-production payroll-pd-releases ioc psp-rtb-priority sbg-payroll-prod mse-commerce-support cx3-production	DONE	Bhat, Chethan	
7	Send Downtime Update to stake holders via Email.	CR, Apps & Downtime details to be shared. Refer ChangeRequests Section. Refer NotificationEmailAddresses Section.	DONE	Bhat, Chethan	

8	Ensure there are no critical batch jobs/RJF jobs running.	Refer JSS View Link. Wait for critical jobs to complete.	DONE	Karra, Vineeth															
9	Bring down below pods 1. Pubsub 2. RJF 3. PAPI 4. PayrollDM	Bring down Pubsub, PAPI (With Maintenance Page) and RJF. Zero Pods With Maintenance Page	DONE	Joshi, Vishwanath															
10	Bring down PSP along with Maintenance Page	Bring Down PSP with Maintenance Page. Zero Pods PSP With Maintenance Page.	DONE	Joshi, Vishwanath															
11	Ensure all apps are down.	Hit Following URLs and make sure all apps are down. <table><tr><th>Application</th><th>Health Check</th></tr><tr><td>PSP</td><td>sh PSPHealthCheck.sh prd</td></tr><tr><td>SAP</td><td>SAP</td></tr><tr><td>PAPI</td><td>PAPI Health Check</td></tr><tr><td>Pubsub</td><td>Pubsub Sanity</td></tr><tr><td>RJF</td><td>RJF Health Check</td></tr><tr><td>PayrollDM</td><td>PayrollDM Health Check</td></tr></table>	Application	Health Check	PSP	sh PSPHealthCheck.sh prd	SAP	SAP	PAPI	PAPI Health Check	Pubsub	Pubsub Sanity	RJF	RJF Health Check	PayrollDM	PayrollDM Health Check	DONE	Singh, Sanket	
Application	Health Check																		
PSP	sh PSPHealthCheck.sh prd																		
SAP	SAP																		
PAPI	PAPI Health Check																		
Pubsub	Pubsub Sanity																		
RJF	RJF Health Check																		
PayrollDM	PayrollDM Health Check																		
12	Update the downtime details section for all the apps.	Update DowntimeDetails Section.	DONE	Bhat, Chethan															
13	Note down All the failed RJF /PSP BatchJobs.	Update Failed/MissedBatchJobs Section.	DONE	Karra, Vineeth															
14	Lock application users on Postgres Hash db.	- Wait for Inflight requests/Running Batch Jobs to complete. - There should not be any connections to Postgres Hash Database. - Lock Postgres Hash Users. <pre>alter user pspapp with nologin; alter user pspadm_owner with nologin; alter user pspbatch_ro_user with nologin; alter user pspbatch_rw_user with nologin; alter user psprijf with nologin;</pre>	DONE	Choubey, Mayank															
15	Ensure there are no active connections to the Postgres Hash Database.	No active connections from PSP / PAPI / Pubsub / RJF / PayrollDM. Monolith : <pre>SELECT datname,username, coalesce (client_hostname, client_addr::text, 'localhost') as client_hostname,count(*) from pg_stat_activity where pid != pg_backend_pid() group by datname,username, coalesce (client_hostname, client_addr::text, 'localhost') order by username;</pre> (Kill Connections if needed) Monolith <pre>select pg_terminate_backend(pid) from pg_stat_activity where pid in (select pid FROM pg_stat_activity where username like '%psp%');</pre>	DONE	Choubey, Mayank															

16	Monitor Postgres Hash Oracle Hash Replication Lag	Wait till Postgres Hash(C2) --> Oracle Hash(B), C2D2, C2D2' Replication lag to zero.	DONE	Choubey, Mayank	
17	GG Replication from Oracle Hash (B) Postgres Hash(C2)	Ensure Extract (B) (for B->C2) stats should not have any table extract stats EPSPHPN2	DONE	Choubey, Mayank	
18	Disable triggers on C2	Host: ggpspadm01.sbg-ppsp-prod.a.intuit.com <code>\i disable_triggers_on_C2.sql</code> Validate triggers are disabled.	DONE	Narlagalla, Praveen Goud	
19	alter sequences start value on B	Alter sequences to start with last_number from C2 +1 ggpspadm01.sbg-ppsp-prod.a.intuit.com <code>\i seq_alter_seq_on_B.sql</code>	DONE	Narlagalla, Praveen Goud	
20	Enable all triggers, FKs on B	Enable all triggers, FKs on B using SETTAG 2 (to exclude going to C2) Validation <code>select constraint_name, table_name, status from dba_constraints where owner='PSPADM' and constraint_type='R' and status='DISABLED';</code> <code>select trigger_name, table_name, status from dba_triggers where owner='PSPADM' and status='DISABLED';</code>	DONE	Choubey, Mayank	
21	Test data extraction from B Extract. Not required as gg_heartbeat job is running on B	Test with insert on heartbeat table to ensure its extracting data from B <code>insert into pspadm.gg_heartbeat values ('B_to_C2', sysdate);</code> stats EPSPHPN2	DONE	Choubey, Mayank	
22	Unlock Oracle Hash Users	Unlock Oracle Hash Users. <code>alter user PSPADM account unlock; alter user PSPAPP account unlock; alter user PSPREAD account unlock; alter user PSPREAD_AWS account unlock; select username, account_status from dba_users where username in ('PSPAPP', 'PSPADM', 'PSPREAD', 'PSPREAD_AWS');</code>	DONE	Choubey, Mayank	

23	Cleanup GG processes for C2->B	Delete extract on C2 <pre>obey ./diroby/dblogin_c2.oby info EPP1PG02 detail stop EPP1PG02 delete EPP1PG02 UNREGISTER EXTRACT EPP1PG02 with database pspagg02</pre> Delete pump on C2 <pre>info ppsphp02 detail stop ppsphp02 delete extract ppsphp02</pre> Delete replicat on B <pre>obey ./diroby/delete_replicats.oby</pre>	DONE	Choubey, Mayank										
24	Validate Production to point to Oracle Hash(A) Database.	Validate Read Write / VMP / RJF Read Instances (3 instances in total). • Pre-requisite: install Stash • Copy .pem file to /tmp folder. • eiamcli login • Validate IDPS Credentials (Database URL/Username/Password) <i>sh Oracle-Hash-A.sh -v</i>	DONE	Bhagat, Suyash										
25	Bring up PSP pods with Maintenance on Oracle	Bring up PSP with Maintenance page. Scale up PSP Pods With Maintenance Page. Ensure correct image is deployed. • Dashboard : Validate from DT Payroll Pod Info • Script : imageDetails.sh prd • Validate database in config map	DONE	Joshi, Vishwanath										
26	Monitor if PSP connecting to Oracle Hash Database	Monitor Database Connection Dashboard and validate.	DONE	Bahl, Pulkit										
27	Ensure there are no errors in the startup logs.	Look for errors in this dashboard Unknown Errors	DONE	Goswami, Harshita										
28	Check the Node level Sanity Logs.	Node Level Sanity Dashboard - Check if all the pods passed sanity tests.	DONE	Bahl, Pulkit										
29	Monitor DB Connections on Oracle Hash.	Monitor db connetions	DONE	Choubey, Mayank										
30	Monitor Overall database /replication health vi CW dashboard	Oracle Hash Database Performance Monitor GoldenGate Replication Dashboard	DONE	Choubey, Mayank										
31	Bring up PSP pods without Maintenance on Oracle Hash.	Bring up PSP without Maintenance page. Scale up PSP Pods Without Maintenance Page. Ensure correct image is deployed. • Dashboard : Validate from DT Payroll Pod Info • Script : imageDetails.sh prd • Validate database in config map	DONE	Joshi, Vishwanath										
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Application	Sanity Link	Status												
PSP	PSP Sanity	OPEN												
SAP	SAP Sanity	OPEN												

33	Monitor DB Performance	We will be using combination of Performance Insights, monitoring dashboard and adhoc scripts / slow query log based on need	DONE	Choubey, Mayank / P opat, Ketan Gunvantrai																
34	Bring up all pods for below services on Oracle 1. PAPI 2. RJF 3. Pubsub 4. PayrollDM	Bring up Pubsub, PAPI and RJF apps without maintenance page. Bring up Apps Without Maintenance Page Ensure correct image is deployed. • Dashboard : Validate from DT Payroll Pod Info • Script : imageDetails.sh prd • Validate database in config map	DONE	Joshi, Vishwanath																
35	Monitor if all apps connecting to Oracle Hash Database	Monitor Database Connection Dashboard and validate.	DONE	Bahl, Pulkit																
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RJF	RJF Unknown Errors	OPEN																		
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37	Run PAPI / Pubsub / RJF / PayrollDM Sanity.	Execute below sanity suites. <table><tr><th>Application</th><th>Sanity Link</th><th>Status</th></tr><tr><td>PAPI</td><td>PAPI Sanity</td><td>OPEN</td></tr><tr><td>Pubsub</td><td>Pubsub Sanity</td><td>OPEN</td></tr><tr><td>RJF</td><td>RJF Sanity</td><td>OPEN</td></tr><tr><td>PayrollDM</td><td>PayrollDM Sanity</td><td>OPEN</td></tr></table>	Application	Sanity Link	Status	PAPI	PAPI Sanity	OPEN	Pubsub	Pubsub Sanity	OPEN	RJF	RJF Sanity	OPEN	PayrollDM	PayrollDM Sanity	OPEN	DONE	Singh, Sanket	
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PayrollDM	PayrollDM Sanity	OPEN																		
38	Run PSP & RJF failed jobs and missed Jobs.	Refer Failed/MissedBatchJobs Section. Run LedgerBalance Job if Missed.	DONE	Karra, Vineeth																
39	Monitor all the apps and databases.	Refer Monitoring Section.	DONE	All																
40	Ensure replication is running without any issues & monitor the lag	Monitor Replication Lag.	DONE	Narlagalla, Praveen Goud																
41	Run Payroll From Production Test company	IKS QBDT PSP Adapters	DONE	Swain, Amrit																
42	Merge generated PRs to environment/prd-usw2-iks to update config map in manifest to point to oracle for PSP Papi Pubsub RJF		DONE	Bhagat, Suyash																
43	Merge PRs to the master branch to update config map to oracle PSP Papi Pubsub RJF refer PR section		DONE	Bhagat, Suyash																

Post Rollback Steps

S No	Task	Steps	Status	Owner	Remarks
1	Send Downtime Update to stake holders via Email.	Refer NotificationEmailAddresses Section.	DONE	Bhat, Chethan	
2	Offline Monitoring for all Apps for 2 hours.	Refer Monitoring Section.	DONE	All	
3	Review & Take action on all the pending Alerts.	Pending Alerts - Critical Alerts PD Pending Alerts - RNO PD	DONE	Kaura, Varun	
4	Review & Close all pending CRs.	Refer ChangeRequests Section.	DONE	Sur, Koustav	
5	Inform partner teams slack channel.	Ask the stake holders to be prepared for the downtime. Inform following Slack Channels. #payroll-pd-releases, #ioc, #psp-rtb-priority, #sbg-payroll-prod, #mse-commerce-support, #cx3-production • payroll-pd-releases • ioc • psp-rtb-priority • sbg-payroll-prod • mse-commerce-support • cx3-production	DONE	Bhat, Chethan	
7	Monitor Critical Batch Jobs.	Monitor Critical Batch Jobs (Required Only for the Cutover) Refer CriticalBatchJobsMonitoring Section.	DONE	Hoolimath, Praveenkumar Karra, Vineeth	
8	Monitor database Stability, Availability & Performance	We will be using combination of Performance Insights, monitoring dashboard and adhoc scripts / slow query log based on need	DONE	Choubey, Mayank (Primary) Popat, Ketan Gunvantrai (Escalation)	
9	Monitor replication from Oracle Hash to Postgres Hash	• Ensure no replication issue	DONE	Choubey, Mayank	
10	Trigger Veridata to validate Delta Data.	Trigger Veridata to validate Delta Data. Veridata Console Link : monolith_rhab_incr_1day	DONE	Unknown User (rrasam) Rajoria, Himanshu	

Monitoring

Apps Health

- Health & Traffic summary shared in the tiptoe slack channel.

Application / Database / Infrastructure Category	Owner	Health			
		Previous Week	Previous Week	Tiptoe	Rollback
		07 Oct 2023	08 Oct 2023	14 Oct 2023	27 Aug 2023
QBDT Adapter	Singh, Sanket	DONE	DONE	OPEN	OPEN
QBDTWS Adapter	Singh, Sanket	DONE	DONE	OPEN	OPEN
DIS Adapter	Singh, Sanket	DONE	DONE	OPEN	OPEN
ADE Adapter	Goswami, Harshita	DONE	DONE	OPEN	OPEN
CDM Adapter	Goswami, Harshita	DONE	DONE	OPEN	OPEN
SAP Adapter	Goswami, Harshita	DONE	DONE	OPEN	OPEN
BRM Adapter	Unknown User (akumari3)	DONE	DONE	OPEN	OPEN
PTC Adapter	Unknown User (akumari3)	DONE	DONE	OPEN	OPEN
Critical Batch Jobs	Hoolimath, Praveenkumar	DONE	DONE	OPEN	OPEN

Batch Jobs during Activity	Karra, Vineeth	DONE	DONE	OPEN	OPEN
Infrastructure Monitoring	Joshi, Vishwanath	DONE	DONE	OPEN	OPEN
PAPI	Bahl, Pulkit	DONE	DONE	OPEN	OPEN
PayrollDM	Bahl, Pulkit	DONE	DONE	OPEN	OPEN
RJF	Karra, Vineeth	DONE	DONE	OPEN	OPEN
Pubsub	Bahl, Pulkit	DONE	DONE	OPEN	OPEN
DB Queries Performance	Sur, Koustav / Rajvanshi, Venkat	DONE	DONE	OPEN	OPEN
Database	Choubey, Mayank	DONE	DONE	OPEN	OPEN
Veridata	Choubey, Mayank	DONE	DONE	OPEN	OPEN
ArgoCd	Joshi, Vishwanath	DONE	DONE	OPEN	OPEN
Unknown Errors	Bhagat, Suyash / Singhal, Vikas	DONE	DONE	OPEN	OPEN
Workflows Performance	Goswami, Harshita / Rajvanshi, Venkat	DONE	DONE	OPEN	OPEN

PSP

S. No	Dashboard	Link	Comments
1	PSP Batch Jobs 1. Successful Jobs 2. Failed Jobs	PSP Batch Jobs	Update JSS Jobs Schedule completed jobs status every 2hours.
2	PSP JSS View 1. Successful Jobs 2. Failed Jobs	PSP JSS View	
3	PSP Service Metrics 1. Response Time 2. No of Requests	PSP Service Metrics	
4	PSP Adapters 1. Response Codes	PSP Adapters	
5	PSP API Gateway 1. TP Numbers (Check - Not Populating) 2. Response Codes	PSP API Gateway	
6	PSP Unknown Exceptions	PSP Unknown Exceptions	
7	Workforce	Workforce	

PAPI

S. No	Dashboard	Link
1	PAPI Gateway 1. TP Numbers (Check - Not Populating) 2. Response Codes	PAPI Gateway
2	PAPI GraphQL Gateway	PAPI GraphQL Gateway

Pubsub

S. No	Dashboard	Link
1	Pubsub (TODO : To enhance to capture all entities)	Pubsub Entity Publish
2	Pubsub TID Analyzer	Pubsub TID Analyzer

Report Job Framework

	Dashboard	Link	Comments
1	RJF JSS View 1. Successful Jobs 2. Failed Jobs	RJF JSS View Splunk Dashboard : QBDT PSP Report Jobs IKS	Update RJF Jobs Schedule completed jobs status every 2hours.
2	Jobs during the tiptoe window	100k_Query ASSISTED_DISCO_CUSTOMER_LIST_NEW Assisted_Agency_Info_Rpt_SIT Back_Dated_Paychecks Duplicate_Adjustments Financial_Ledger Missed_Payments_High_Priority Psp_Cancelled_Assisted_Info Assisted_Activation_Report Assisted_Agency_Info_Rpt_Others Assisted_Agency_Info_Rpt_SUI Assisted_Agency_aid_rpt_SIT Assisted_Agency_aid_rpt_SUI Assisted_On_Hold_Info Back_Dated_voids Bad_EE_Accounts Bad_Funding_Account Cancelled_By_MissedPayrollsBatchJob EU_activ_err_count_exceed_limit Eu_Err_Aging_Info Financial_Transaction Foreign_Routing_Number Missed_payments_not_immediate New_Signups_Report mmt_rec_count_exceed PSP_COMPANY PSP_Entitlement_AppUser PSP_STATEID PSP_contact dd_limit_inf_jirapsp_6681 Bank_Report Complia_Adjustments	

Performance

Traffic	Category	Dashboards
OLTP	Workflow Performance	OLTP Workflow Performance Comparison
	Query Performance	OLTP Query Performance Comparison
	Query Analysis	Query Analysis
	SAP Performance	SAP Query Performance Comparison
	Papi Workflow Performance	Comparison QBDT Papi Metrics Prod
	VMP Workflow Performance	Comparison QBDT VMP Metrics Prod
	PSP Service Metric	PSP Service Metrics
	Pubsub workflow performance	Pubsub Metrics Comparison
OLAP	Batch Job Performance	Batch Jobs Performance Comparison
	Query Performance	Batch Jobs Query Performance Comparison

Tiptoe Batch Jobs Monitoring

S No	Jobs	Validation Scripts
1	AMO	
2	Entity Event	
3	EmailGateway	
4	EMSBSToBRMDDataSyncProce ssor	
5	IAMEmailAddressProcessor	

6	CheckPrint	
7	EntitlementProcessor	
8	QbdtUnprocessedRequestsRetry	
9	WorkersCompProcessor	
10	AssistedUsageDataSyncProcessor	
11	Ledger Balance	

BatchJobs in Activity Window

S. No	Jobs	Comments
1.	SendCustomEmailsProcessor SoxReportProcessor ScheduledEmailsProcessor AccountServiceSyncExceptionProcessor LedgerBalanceProcessor StateReportProcessor EntityEventRetryProcessor BulkWorkforceInviteProcessor MtlTransactionReportEnrichProcessor EVSCompanyProcessor RealTimeEntityEventRetryProcessor RTBAutomationProcessor IamEmailAddressProcessor WorkersCompProcessor CheckPrintProcessor PSPTToEMSBSDDataSyncProcessor EmailGatewayProcessor AssistedUsageDataSyncProcessor RetryQbdtUnprocessedRequestProcessor EntityEventProcessor LedgerOperationsProcessor AMOMessageProcessor EntitlementProcessor CostCoPISqlJobsProcessor EDRAssociationFixPISqlJobsProcessor EFTPSOnHoldPaymentPISqlJobsProcessor NCDFixPISqlJobsProcessor PayrollFraudBatchPurgePISqlJobsProcessor RetryEntitlementActivationPISqlJobsProcessor ValidateEmployeeWagePlansPISqlJobsProcessor	

Critical Batch Jobs Monitoring

S. No	Steps	Jobs
1	Monitor Critical Batch Jobs.	- EFTPS All - Ledger Balance - EmailGateway - EMSBSToBRMDataSyncProcessor - CheckPrint - EntitlementProcessor - QbdtUnprocessedRequestsRetry - WorkersCompProcessor - AssistedUsageDataSyncProcessor

Postgres Hash Specific Monitoring

S.No	Dashboard	Comments
1	DT Payroll Oracle Hash	
2		

3		
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Workflows Monitoring

S. No	Workflow/AppName	Validation Points
1	Run Payroll	Hoolimath, Praveenkumar will run from test company.
2	SAP Monitoring	Singhal, Vikas to update script here
3	QBDT/QBDTWS Monitoring	Swain, Amrit to update script here

DB Queries Monitoring

S. No	Dashboard	Link
1	PSP DB Telemetry	Monitor long running queries. PSP DB Telemetry

Database Monitoring

S.No.	Dashboard	Link
1	Monitor Postgres Hash --> Oracle Hash Replication Lag	GoldenGate Replication Dashboard
2	Ensure Veridata validation is green from Postgres Hash to Oracle Hash	veridata console
3	Monitor Connections and Long Running Queries	
4	Postgres Hash Specific Dashboards	
5	Postgres Hash database dashboard	Choubey, Mayank to update

Infrastructure Monitoring

Golden Gate Signals

Application	Module	Golden Gate Signals Link	Owner
PSP	QBDT Adapter	QBDT Adapter Golden Signals	Bhagat, Suyash
	ADE Adapter	ADE Adapter Golden Signals	Bhagat, Suyash
	BRM Adapter	BRM Adapter Golden Signals	Bhagat, Suyash
	DIS Adapter	DIS Adapter Golden Signals	Bhagat, Suyash
	EWS Adapter	EWS Adapter Golden Signals	Bhagat, Suyash
	PTC Adapter	PTC Adapter Golden Signals	Bhagat, Suyash
	CDM Adapter	CDM Adapter Golden Signals	Bhagat, Suyash
	Batch Jobs	Batch Jobs Golden Signals	Bhagat, Suyash
	OLTP query having tp95>0.2	OLTP Query performance	Goswami, Harshita
PAPI	PAPI	PAPI Golden Signals	Rajoria, Himanshu
RJF	RJF	RJF Golden Signals	Joshi, Vishwanath

Wavefront

Application	Wavefront Monitor	Owner
PSP	PSP Wavefront Monitor	Bhagat, Suyash
PAPI	PAPI Wavefront Monitor	Rajoria, Himanshu

Pubsub	Pubsub Wavefront Monitor	Rajvanshi, Venkat
RJF	RJF Wavefront Monitor	Joshi, Vishwanath

AppDynamics

This is the backup for Infrastructure monitoring.

Environment	App Dynamics Link	Owner
Production	DTPayroll AppDynamics BaseURL	Goswami, Harshita

Failed/Missed Batch Jobs

Validation queries for the jobs in the tiptoe window can be found [here](#)

App	Category	Job Names / Screenshots (Tiptoe)	Job Names / Screenshots (Rollback)
Batch Jobs	Scheduled jobs	SendCustomEmailsProcessor SoxReportProcessor ScheduledEmailsProcessor AccountServiceSyncExceptionProcessor LedgerBalanceProcessor StateReportProcessor EntityEventRetryProcessor BulkWorkforceInviteProcessor MtlTransactionReportEnrichProcessor EVSCompanyProcessor RealTimeEntityEventRetryProcessor RTBAutomationProcessor IamEmailAddressProcessor WorkersCompProcessor CheckPrintProcessor PSPTToEMSBSDDataSyncProcessor EmailGatewayProcessor AssistedUsageDataSyncProcessor RetryQbdtUnprocessedRequestProcessor EntityEventProcessor LedgerOperationsProcessor AMOMessageProcessor EntitlementProcessor CostCoPISqlJobsProcessor EDRAssociationFixPISqlJobsProcessor EFTPSOnHoldPaymentPISqlJobsProcessor NCDFixPISqlJobsProcessor PayrollFraudBatchPurgePISqlJobsProcessor RetryEntitlementActivationPISqlJobsProcessor ValidateEmployeeWagePlansPISqlJobsProcessor	
	Missed Jobs		
	Failed Jobs		

RJF	Scheduled jobs	100k_Query ASSISTED_DISCO_CUSTOMER_LIST_NEW Assisted_Agency_Info_Rpt_SIT Back_Dated_Paychecks Duplicate_Adjustments Financial_Ledger Missed_Payments_High_Priority Psp_Cancelled_Assisted_Info Assisted_Activation_Report Assisted_Agency_Info_Rpt_Others Assisted_Agency_Info_Rpt_SUI Assisted_Agency_aid_rpt_SIT Assisted_Agency_aid_rpt_SUI Assisted_On_Hold_Info Back_Dated_voids Bad_EE_Accounts Bad_Funding_Account Cancelled_By_MissedPayrollsBatchJob EU_activ_err_count_exceed_limit Eu_Err_Aging_Info Financial_Transaction Foreign_Routing_Number Missed_payments_not_immediate New_Signups_Report mmt_rec_count_exceed PSP_COMPANY PSP_Entitlement_AppUser PSP_STATEID PSP_contact dd_limit_inf_jirapsp_6681 Bank_Report Complia_Adjustments	
	Missed Jobs		
	Failed Jobs		